UNILAK RWAMAGANA CAMPUS

SOFTWARE ENGINEERING AND USER CENTRED DESIGN

Citizen Issue Reporting and Escalation System

CIRES

Milestone 1

Discovery Report

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PART I THE COMMUNITY REPORTING PROBLEM

A Purpose of the enquiry

CIRES is a proposed Citizen Issue Reporting and Escalation System for following local road and drainage reports from submission to a recorded outcome. The project begins with the question of what happens after a resident reports a community problem. A concern may be received by an office, passed to an officer and discussed again during follow-up. The proposed investigation examines whether the people involved can identify the report, see who is responsible and understand what action has been taken.

The approved topic proposal identified three possible gaps in reporting, ownership and follow-up. These gaps provide a starting hypothesis for discovery. They do not demonstrate how frequently a problem occurs in a particular locality. The supplied presentation contains an illustrative process analysis and fictional interview responses. This report develops their meaning while retaining their evidence status. No completed local interviews or measured service delays were documented in the material.

The pilot boundary is one area, two issue categories and three operational roles. Roads and drainage provide concrete examples that residents and responsible staff can describe. The proposed users are residents, assigned officers and a supervisor. This boundary supports a manageable enquiry into the reporting trail before the team makes detailed design decisions.

B Team contribution

NKURUNZIZA James and UWASE Phoebe form the project team. The supplied cover identifies their names and registration numbers without allocating separate lead roles. Both members therefore remain jointly responsible for explaining the problem, preparing the incident questions, organising the scenario evidence and documenting the proposed pilot. The team must confirm any division of responsibilities when maintaining the repository and planning the subsequent milestones.

C Why the issue deserves investigation

The first discovery criterion concerns reality. A specific incident is needed to show which office received a report, how it was recorded and what happened next. The present proposal points to a plausible reporting difficulty, while documented user accounts are still needed to confirm its local form. The second criterion concerns frequency. Comparing separate

accounts would show whether the same gap occurs repeatedly. The source material supplies no count or proportion, so frequency remains a research question.

The third criterion concerns the burden on the person affected. A resident may need to repeat a location or contact an office again. An officer may have to reconstruct the original description before acting. A supervisor may need to enquire about ownership before addressing a delay. These are possible consequences illustrated in the walkthroughs. Their duration, cost and prevalence have not been measured.

Reachability concerns whether people with relevant experience can be consulted. A resident, an assigned officer and a supervisor can each explain a different stage of the same report. Buildability concerns whether software can address a limited part of that journey within the course. A single report record, a clear assignment and one overdue escalation define a focused opportunity, subject to validation of how the local reporting process operates.

PART II CONTEXT AND PARTICIPANTS

A The reporting journey

The blocked-drain example begins when a resident notices water collecting near a home and identifies a nearby landmark. The resident contacts a local office and describes the concern. An officer determines whether the issue falls within the area and who should take action. The resident later seeks an update. Each handoff can preserve or lose useful information, including the original description, the location, the expected response and the latest action.

The discovery question is which stage creates the longest or most uncertain gap and what record is available at that point. A report reference would support later identification, but its value depends on whether staff can retrieve the same case. A deadline would support follow-up, but it must have an owner and a meaningful action record. The enquiry therefore considers the relationships between reporting, assignment, visibility and closure.

B Stakeholder responsibilities

The resident is the person who reports an issue and needs an understandable account of progress. The assigned officer needs enough information to locate the issue, accept responsibility and record work undertaken. The supervisor needs visibility of the assigned workload and overdue open cases. Their perspectives differ, but all three depend on the same report history.

The wider stakeholder group includes pilot-area leadership, local office receiving staff, maintenance personnel and residents affected by the road or drainage condition. Leadership may influence authority and operating rules. Receiving staff may determine whether the original description is usable. Maintenance personnel may contribute action information, even if they do not hold a separate application role. Residents outside the pilot boundary remain relevant to the wider service context, while the first release focuses on the defined area.

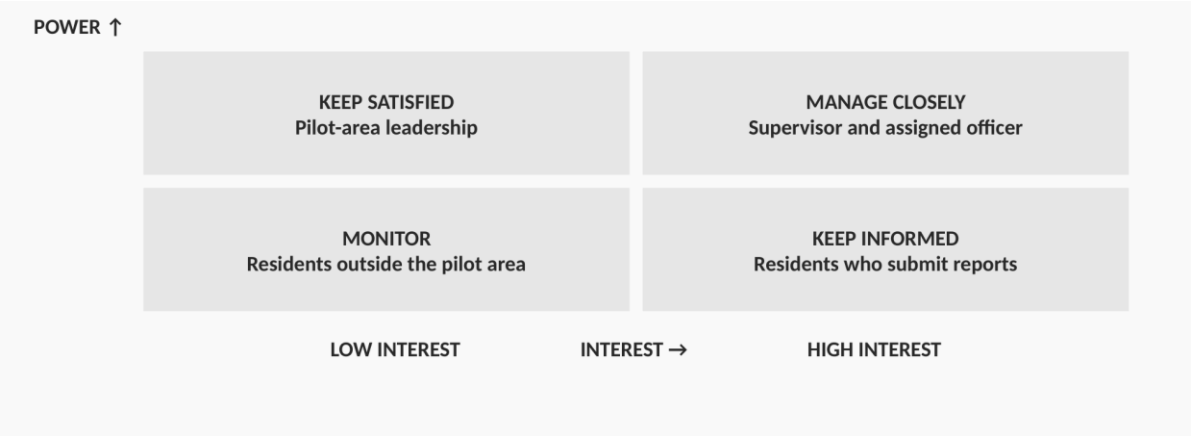


Figure A Stakeholder power and interest map from the supplied presentation

The map assigns proposed levels of power and interest to the identified groups. It places the supervisor and assigned officer in the group requiring close involvement because their responsibilities determine whether the trail can be maintained. Residents who submit reports need clear information about progress. The positions represent proposed pilot responsibilities and require confirmation through the local enquiry.

C Existing reporting routes

An office visit allows a resident to describe an issue directly and answer immediate questions. Its usefulness for later follow-up depends on how the receiving office records the concern and provides a reference. A phone call or message can reduce the need for travel but may leave the description and later decisions in separate conversations. A paper register can preserve receipt details, while retrieval and updates depend on the way staff maintain it. These comparisons describe the routes considered in the presentation and require local observation.

Published complaint-handling information provides relevant context. RURA describes receiving complaints through several channels and explains procedures for recording an oral complaint in writing (Rwanda Utilities Regulatory Authority, n.d.). The Office of the

Ombudsman describes receiving complaints related to injustice and examining how administrative organs resolve them (Office of the Ombudsman, n.d.). These examples show that receipt, recording and follow-up are recognised parts of complaint handling. Their mandates differ from the proposed roads-and-drainage pilot, and they do not establish local CIRES performance.

PART III DISCOVERY METHOD

A Incident questions and recording approach

The proposed enquiry uses three conversations covering the resident, officer and supervisor viewpoints. The resident is asked to walk through a recent report and explain where the process started and what response followed. The officer is asked to describe the last issue received and how responsibility was established. The supervisor is asked to recall an overdue issue and identify what first brought it to their attention. These incident questions follow the lecturer's emphasis on examining behaviour.

Each incident record should preserve the reporting channel, issue category, first response, transfers, follow-up effort and outcome. Follow-up questions examine what showed that a report had been received, what information was missing and what record demonstrated that action was overdue. These details support later comparison between accounts and help distinguish an unclear workflow from an isolated misunderstanding.

Participation should be voluntary and notes should use participant codes. Permission is needed before recording a discussion. Unnecessary identifiers, personal contacts and sensitive complaint details should be excluded from shared research records. The supplied presentation does not document completed consent or interview sessions. The following dialogues remain fictional walkthroughs prepared for design exploration.

B Resident walkthrough

The resident scenario concerns a blocked drain near a home. The opening question asks, "Take me back to when you noticed the blockage. What did you do?" The illustrative response states, "I told the local office where the water was collecting and described the nearest landmark." The answer establishes the information the resident tried to provide and raises a question about whether it was preserved in a shared record.

The next question asks, “How did you know whether your report had reached the right person?” The response states, “I had no reference. When I called again, I had to explain the location from the beginning.” This illustrates the possible connection between missing acknowledgement and repeated explanation. It suggests that the enquiry should establish whether residents receive a reference and whether receiving staff can retrieve the earlier account.

The final question asks, “What update would have helped you decide whether to call again?” The response is, “A message showing who received the report and when I could expect another update.” This answer expresses a possible information need. It should be checked against the participant’s actual behaviour before selecting a notification channel or promising an automatic response time.

C Assigned officer walkthrough

The officer scenario begins with a drainage report arriving at an office. The question, “What would you check before accepting this report?” receives the illustrative response, “I would check whether it falls within our area and whether the location is clear enough to find.” The scenario identifies the service boundary and location quality as possible conditions for assignment.

The follow-up asks, “Tell me about the handoff. What might the next officer need?” The response states, “The original description, the location, the deadline and a note about any action already taken.” This identifies the information that should survive transfer. It also suggests that assignment cannot be understood simply as a name appearing on a list. The responsible person needs the context and expectations attached to the report.

The final question asks, “What would you record before closing it?” The response is, “The work completed and the date. The resident should be able to see a short explanation.” This illustrates why closure needs an action summary. A status change alone may not explain what happened. The proposed enquiry must determine who records the work and which details can appropriately be shown to the resident.

D Supervisor walkthrough

The supervisor scenario concerns a report that has passed its deadline. The opening question asks, “How would you first notice that action was overdue?” The response states, “I would check the open reports for any deadline that had passed without a resolution.” This links the

concept of delay to an open status and a due time. It does not establish an agreed local response standard.

The next question asks, “What information would help you decide what to do next?” The response is, “The assigned officer, the last action, the due time and how long the report has been overdue.” The answer illustrates the value of a connected history for supervisory follow-up. A deadline without context may identify lateness but still leave the supervisor unable to decide whom to contact or what has already been attempted.

The closing question asks, “What would make the escalation useful?” The response states, “One clear entry in my queue with a timestamp, so I can follow up without duplicate alerts.” This illustrates the proposed single-escalation rule. Actual user review is needed to confirm that this is sufficient for the pilot and to determine how supervisors will follow up after viewing the entry.

PART IV INTERPRETATION OF THE SCENARIOS



Figure B Affinity map of reporting risks from the supplied presentation

A Five candidate difficulties

The first concern is reporting receipt. The resident’s statement, “I had no reference,” illustrates the absence of a convenient way to identify the previous contact. A receipt record could help connect later enquiries to the original report. The second concern is ownership. The officer’s reference to “the original description, the location, the deadline” illustrates the information needed to accept and continue responsibility. A case record needs both an assigned officer and usable details.

The third concern is progress visibility. The resident wanted to know who received the issue and when another update could be expected. This suggests a need to understand the current stage without reconstructing the whole conversation. The fourth concern is overdue visibility. The supervisor's statement about checking "open reports for any deadline that had passed" identifies a possible need to detect delay from the stored report rather than wait for another resident enquiry.

The fifth concern is meaningful closure. The officer's proposed record of "the work completed and the date" illustrates why a final status should have an explanation. These five concerns remain candidates derived from fictional responses. The affinity map organises them for validation, while the operating enquiry must establish which difficulties occur in actual incidents.

B User profiles for the pilot

The resident profile centres on making a report and understanding what happens next. The person needs to describe a location clearly, obtain a reference and see an understandable update. The supplied scenario highlights the frustration of repeating the same description. The profile does not assign an invented age, occupation or device because those characteristics were not established through field evidence.

The assigned officer profile centres on accepting a clearly defined case and showing progress through to an explained outcome. The supervisor profile centres on finding overdue work with sufficient context to follow up. The two staff profiles share a need for reliable assignment and history, while their authority differs. These provisional profiles will need to be refined through actual accounts of the reporting process.

C Brainstorming the opportunity

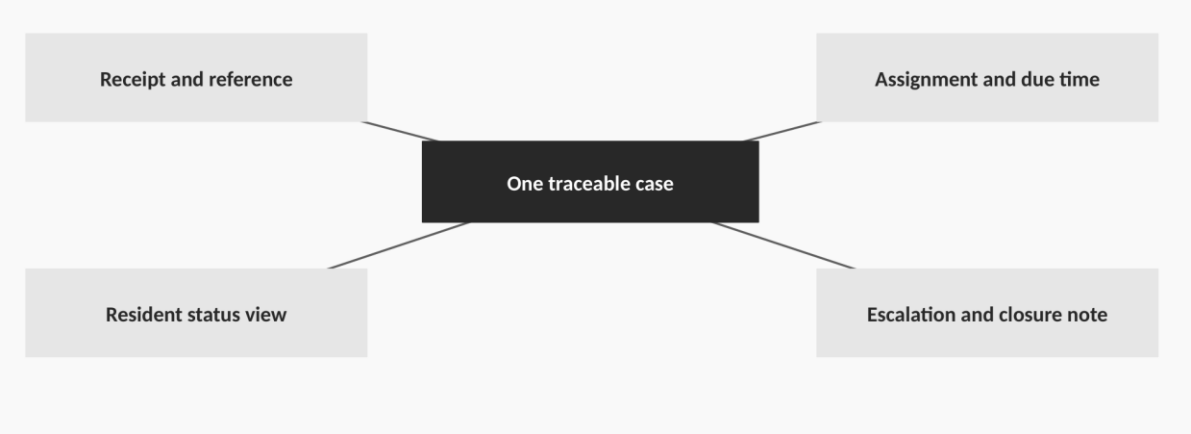


Figure C Brainstorming the report trail from the supplied presentation

The brainstorming exercise brings receipt, assignment, status visibility and escalation together under a single traceable case. The proposal is that the report should retain the information needed by the next participant. A shared record would connect the resident's original concern to an officer's action and a supervisor's follow-up. This direction remains a proposed response to the scenario concerns.

PART V PROBLEM STATEMENT AND PROJECT DIRECTION

Residents who report local road or drainage issues may struggle to tell whether their concern was received, assigned or acted on. If the reporting route provides no shared reference or status record, residents may need to contact the office again and supervisors may learn about delays late. The proposed problem is therefore a gap in the visibility and continuity of the report trail.

The approved topic proposal and illustrative analysis provide the basis for this working statement. The material does not contain observed failure rates, measured response times or completed local interview records. The next research step must establish the actual reporting route, responsibility for each stage and the nature of any delay before these concerns are treated as confirmed findings.

The proposed pilot will receive road and drainage reports within one area. It will identify each report, assign it to an officer with a configurable deadline, display its progress and flag an overdue open case to the supervisor once. Closure will require a record of the action taken. This scope supports a focused course project and provides a clear basis for the requirements milestone.

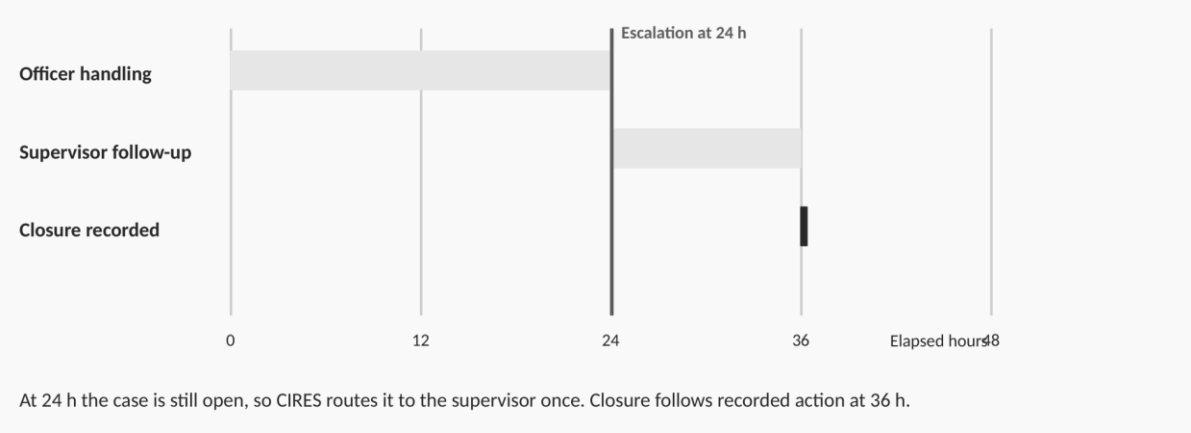


Figure D Illustrative timing chart retained in monochrome

The timing chart shows an example in which a report is still open at 24 hours and receives one escalation. Closure follows at 36 hours after action is recorded. These values illustrate the sequence and remain configurable. They are neither measured service results nor an approved response standard for a public office.

Milestone 1 has established a structured discovery baseline around three roles and a limited reporting journey. The next milestone must convert the proposed workflow into testable requirements while checking it with real users. The final discovery evidence should include anonymised incident accounts, a comparison of local reporting routes and confirmation of the pilot responsibilities.

References

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